

**Network Segmentation Through VLAN Implementation and Inter-VLAN Routing:
A Solution to Flat Network Vulnerabilities in Enterprise Environments**

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Abstract

Flat network architectures present a real and often underestimated risk to enterprise environments. When every device shares a single broadcast domain, one compromised machine can reach everything else on the network—no boundary, no checkpoint, nothing in between. This paper examines that problem and makes the case for Virtual Local Area Network (VLAN) segmentation combined with inter-VLAN routing as the most practical solution available. The focus is on both the security advantages and the operational realities of deploying VLANs in a real environment, including two common implementation approaches: router-on-a-stick and multilayer switching. The conclusion is straightforward—segmented networks are more secure, more manageable, and better suited for growth, and the tools to build them have been available and well-tested for decades.

Keywords: VLAN, inter-VLAN routing, network segmentation, broadcast domain, flat network

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Introduction

Networks get complicated fast. What started as a handful of computers sharing a printer in an office has turned, in many organizations, into something almost unrecognizable—hundreds or thousands of endpoints spread across multiple floors or buildings, all running voice, video, and data simultaneously. And somewhere along the way, a lot of those networks never really got redesigned. They just grew.

The result is what network engineers call a flat network. In a flat network, every device belongs to the same broadcast domain. That means every device sees every broadcast sent by every other device. It also means a machine infected with malware has no real barrier between itself and the rest of the network. No segmentation. No routing boundary. Just open access.

This paper argues that VLAN segmentation, combined with properly configured inter-VLAN routing, directly addresses the two most significant weaknesses in flat network design: security exposure and performance degradation. VLANs aren't a new technology—they've been around since the 1990s—but a surprising number of organizations still haven't implemented them fully. The reasons vary, but the most common one is probably that nothing has gone catastrophically wrong yet. That's not a good enough reason to wait.

The Problem: Flat Network Architecture and Its Limitations

Security vulnerabilities There's a principle in cybersecurity called least privilege—the idea that any user, system, or process should have access to only what it actually needs and nothing more. Flat networks make that principle nearly impossible to enforce.

When a device connects to a flat network, it becomes a peer to every other device on the segment. From a Layer 2 perspective, all it needs is a working NIC and a valid MAC address. No credentials required, no restrictions applied. According to Lammle (2020), this kind of unrestricted access is precisely what allows an attacker to conduct ARP poisoning and man-in-the-middle attacks from a single compromised endpoint, without ever crossing a routing boundary (p. 78). No privilege escalation. No firewall bypass. Just the right packets.

The WannaCry ransomware outbreak in 2017 showed exactly how bad this can get. The malware used an SMB exploit to hop from machine to machine, and in organizations running flat networks, it spread almost without resistance. Odom et al. point out that lateral movement across uncontrolled broadcast domains is one of the most consistently exploited vectors in enterprise breach scenarios (Odom et al., p. 214). In a properly segmented network, the infection would have been contained to the VLAN where it started. That's a fundamentally different outcome.

There's also the issue of passive reconnaissance. On a flat network, an attacker who has compromised any device can run a network scan and see everything—servers, workstations,

printers, cameras, whatever is on the segment. That kind of visibility is enormously useful to an attacker and completely unnecessary in a well-designed network.

Performance Limitations The performance argument for VLANs is a bit more technical but just as important. Every broadcast frame on a network gets delivered to every device in the broadcast domain. Every single one. In a network with 400 endpoints, a single ARP request hits all 400 machines.

Now multiply that across the dozens or hundreds of broadcasts per second a busy network generates. The devices spend real CPU cycles processing frames that have nothing to do with them. Bandwidth gets consumed before any useful work happens. In smaller environments this overhead is manageable, but as the network scales, the cost of shared broadcast domains grows proportionally and there isn't really a way around it without introducing logical segmentation.

VoIP traffic is where this shows up most visibly. Voice is extremely sensitive to latency—a 150ms delay in a voice packet becomes an audible glitch in a call. In a congested flat network, that kind of delay is common. Quality of service configurations can help at the margins, but the underlying problem is a shared broadcast domain that can't prioritize traffic at the Layer 2 level.

Scalability Constraints The 802.1D Spanning Tree Protocol adds yet another layer of difficulty. STP was designed for relatively small networks, and in a large flat environment,

topology-change notifications flood the entire domain any time a link state changes—even something as routine as a port going up or down. Convergence times under classic STP can reach 30 to 50 seconds, which is a long time for users to be without connectivity. The bigger the flat network, the more painful this becomes.

The Solution: VLANs and Inter-VLAN Routing

Virtual Local Area Network Fundamentals A VLAN is, at its core, a logical way to divide a physical network into separate broadcast domains without adding extra switches or cabling. The IEEE 802.1Q standard handles this by inserting a four-byte tag into each Ethernet frame that identifies which VLAN it belongs to. Devices in VLAN 10 don't see broadcasts from VLAN 20. From each device's perspective, the other VLANs basically don't exist.

That boundary matters enormously from a security standpoint. A compromised machine in the guest VLAN can't ARP its way into the finance VLAN—it would have to cross a routing boundary first, and that routing boundary is where access controls live. The attacker's options are immediately narrowed to whatever is on the same VLAN as the initial infection.

There's also a practical administrative benefit that's easy to overlook. VLANs let administrators group devices by function regardless of physical location. If a user moves to a different floor, there's no need to recable anything—just reassign the switchport to the correct VLAN and the user is back in the right network segment. In a large environment, that kind of flexibility adds up.

Inter-VLAN Routing Segmenting the network with VLANs is the first step, but most environments also need devices in different VLANs to communicate with each other. An HR workstation in VLAN 20 might need access to a database server in VLAN 30. A guest device might need to reach the internet but nothing internal. Inter-VLAN routing is how those decisions get made.

The first approach is what's commonly called router-on-a-stick. A single router connects to the switch through a trunk link and handles inter-VLAN traffic using logical subinterfaces, one per VLAN. It's relatively simple to configure and doesn't require additional hardware—Packet Tracer lab exercises demonstrate the basic setup pretty clearly. The catch is that all inter-VLAN traffic has to funnel through a single physical uplink, which creates a bottleneck as traffic volumes grow.

The second approach uses a multilayer switch, and this is what most enterprise environments should be running. Multilayer switches perform Layer 3 routing in hardware using Switched Virtual Interfaces (SVIs). Each VLAN gets an SVI that acts as its default gateway, and routing between VLANs happens inside the switch at wire speed—no external router required. For anything beyond a small office, this is the right tool (Cisco Systems, 2019).

Spanning Tree and VLAN Design With VLANs in place, Spanning Tree Protocol becomes much more manageable. Instead of a single STP topology covering the entire network, each VLAN maintains its own instance. Cisco's PVST+ and Rapid PVST+ make this possible,

and the performance difference is significant. Under Rapid PVST+, convergence drops from that 30-50 second window down to under a second in most conditions (Tanenbaum & Feamster, 2022). It also allows different VLANs to use different root bridges, which enables load balancing across redundant uplinks—something a flat network simply can't do.

Integrating Access Control lists

VLANs create the walls. Access Control Lists define the doors—what can pass through, in which direction, and under what conditions. Without ACLs, inter-VLAN routing is just routing: traffic moves between segments freely and the security benefit of segmentation is mostly lost.

Extended ACLs applied at the inter-VLAN routing layer give administrators granular control over cross-segment traffic. A guest VLAN can be restricted to internet-only access. Workstation VLANs can be blocked from directly accessing server management ports. Finance servers can be isolated from everything except specific permitted sources. The policies are enforced before traffic reaches its destination, which means there's no path for a guest device to reach internal resources unless a rule explicitly allows it. Combined with VLAN segmentation, this creates a layered defense that's considerably harder to work around than anything a flat network can offer.

Implementation Considerations

There are a few things that commonly go wrong during VLAN deployments, and it's worth being direct about them. The most well-known is native VLAN misconfiguration. VLAN 1 is the default native VLAN on Cisco switches, and leaving it as-is opens the network to VLAN hopping attacks. In a double-tagging attack, an attacker crafts frames that carry two 802.1Q tags—the outer one matching the native VLAN, the inner one targeting another VLAN. The switch strips the outer tag and forwards the frame into the target VLAN. The fix is simple: reassign the native VLAN to an unused ID on all trunk ports (Lammle, 2020). It gets overlooked more often than it should.

Dynamic Trunking Protocol (DTP) is another issue. DTP is enabled by default on most Cisco interfaces, which means the switch will automatically try to negotiate trunk mode with connected devices. If a rogue device connects and successfully negotiates a trunk, it gains access to every VLAN on the switch. Disabling DTP and explicitly setting each port as either access or trunk isn't optional in a production environment. It's basic hygiene.

Documentation is something a lot of teams push off until something breaks and nobody knows what VLAN 47 is for or who owns it. VLAN databases grow over time—sometimes organically, sometimes chaotically—and without a maintained register mapping VLAN IDs to names, subnets, and responsible teams, troubleshooting becomes a guessing game. It doesn't have to be elaborate. A spreadsheet is fine. But it has to be kept current.

Finally, the migration itself. Going from a flat to a segmented network rarely happens in a single maintenance window. A phased approach is much safer—bring the management VLAN

online first to establish a stable out-of-band control path, then move non-critical workloads, then critical ones. Odom & Healy (2021) specifically recommend this sequencing, noting that a dedicated management VLAN gives administrators reliable access throughout the migration even if something goes wrong with a production VLAN cutover (p. 331). That's good advice.

Conclusion

A lot of organizations know they should segment their networks. The arguments have been in the literature for years. And yet flat networks are still common, usually because the disruption of a migration doesn't seem worth it when everything is technically working.

The problem is that technically working and actually secure are not the same thing. A flat network keeps users connected and traffic flowing, but it also means a single phishing email or a misplaced USB drive could give an attacker a clear path to every device on the network. That's a risk posture that doesn't hold up in 2026.

VLANs aren't a perfect solution—nothing in networking is—but they are the right solution for this specific problem. The technology is mature, the hardware supports it, and the implementation path is well documented. The challenges covered in this paper are real but manageable. And the alternative—waiting until something goes wrong—keeps getting more expensive every year.

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